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Original Research

Relationship between Neck Musculature and Head Kinematics during a Collision in University American Football Players

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ABSTRACT

OBJECTIVES Sport-related concussion (SRC) has been a focus of recent attention. Amongst American football players, it is often the result of high-energy collisions that cause the head to undergo high kinematics of linear acceleration (LA) and angular acceleration (AA). A player generally develops head and neck stiffness by contracting the neck muscles during a collision. Thus, the head kinematics experienced by a player during a collision may be related to the strength of these muscles.

METHODS In a sample of 22 players from T University in the Kanto collegiate American football association second division, we measured neck strength, using a handheld dynamometer, Micro FET (Nihon Medix Co., Ltd.), and neck circumference, using a measure. We then measured their head kinematics (LA and AA) in collisions during seven competition games using an impact-sensing mouthguard. We performed Spearman's rank correlation to assess the relationships between head kinematics, neck circumference, and neck strength.

RESULTS A significant negative correlation between Top10%LA and neck flexor strength was observed during a collision. Additionally, significant negative correlations were observed between head kinematics and neck circumference.

CONCLUSIONS Our main finding was that a stronger and thicker neck may reduce head kinematics during a collision. Although neck strength seems to be protective against SRC in collisions, further research is needed into the involvement of individual muscles and timing of muscle contraction. In our discussion, we emphasize the importance of neck strength training for minimizing SRC.

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Introduction

In American football, intense collisions make the sport attractive to an audience. However, repeated head impacts carry a high risk of sport-related concussion (SRC), which is the most notable current health issue in sports [1,2]. For the past 5 years, we have collected data regarding head kinematics during collisions in Japanese American football teams. We reported that there is a positive correlation at the time of collision between peak head linear acceleration (LA) and peak head angular acceleration (AA) [3]. This indicates that substantial LA is experienced as the head rotates. Two factors are generally believed to be critical to head kinematics during a collision. First, the location of impact is important because, even at same magnitude of LA, different AA results from impacts relative to the head's center of gravity [4]. Second, neck strength is important because, in collisions, players stiffen the head and neck by contracting the neck muscles, resulting in trunk rigidity [5-

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7]. Because bigger and stronger primary muscles generally help increase rigidity of a joint and muscle [8], neck strength training has been indicated as an efficient preventive method to reduce a risk of SRC [5-7]. The National Football League head, neck and spine committee reported that increased spring stiffness in the neck of a Hybrid Ⅲ dummy reduced the concussion threshold by 35% [9]. Additionally, a mathematical analysis reported that greater neck stiffness reduces head kinematics, suggesting that greater neck strength may prevent SRC [10]. Conversely, several in vivo studies have not shown reduced head kinematics in players with greater neck strength [5,7,11]. It has been reported that players with stronger upper trapezius muscles experience greater head kinematics during collisions [12].

In clarifying the mechanisms of SRC in American football collisions, neck strength and head kinematics must be considered. This study aimed to examine whether greater neck strength contributes to reduced head kinematics during collisions.

Methods

Subjects

This study's subjects were 22 players of T University in the Kanto university American football association second division. The subjects' physical characteristics are shown as follows: age; 21.3±1.4y.o., height; 173.5±5.4cm, weight; 82.5±11.7kg). Neck strength and circumference measurements were performed prior to the competition season. We excluded players who had difficulties in measuring neck strength by an injury. All subjects received a thorough explanation of the study's purpose and consented to participate. The study was conducted with the approval of the Tsukuba University Sport Theory Committee (Approval No.: 23-93).

Neck Strength Measurement

A handheld dynamometer was used to measure isometric strength of the neck (Micro FET; Nihon Medix, Chiba, Japan) <Figure 1>. The measurement was taken one month prior to a regular season of competition. With the neck in a neutral position in lying on the table and an assistant stabilizing the trunk, an JSPO-AT (Japan Sports Association-athletic trainer) who used to measure it performed three measurements in four directions (flexion; supine, extension; prone, and left

Figure 1. A handheld dynamometer, Micro FET was used to measure isometric strength of the neck.

Figure 2. Neck strength measurement. Left; neck flexion, Middle; neck lateral flexion, Right; neck extension.

Figure 3. Neck circumference was measured with a measure below the thyroid cartilage.

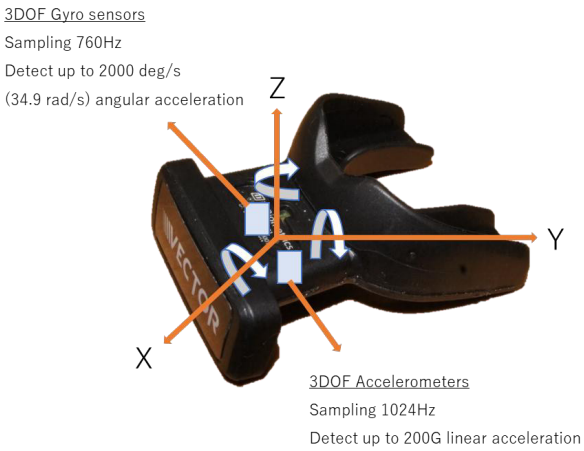

Figure 4. A mouthguard equipped with 6DOF.

and right lateral flexion; side lying) <Figure 2>. A JSPO-AT resisted by the equivalent force applied while the subjects were instructed to exert their maximal neck strength. The ICCs of the measurements are 0.99 in flexion. The average values were calculated.

Neck Circumference Measurement

Neck circumference below the thyroid cartilage was measured twice with a tape measure by a JSPO-AT <Figure 3>. The measurement was taken one month prior to a regular season of competition. The subjects sat comfortably on the table with their hands placed on the thigh. The average value was calculated.

Head Kinematics Measurement

Measurement Method

An impact sensor mouthguard (Vector i1; Biometrics, Wallingford, CT, USA) with six degrees of freedom (a three-axis acceleration sensor and a three-axis gyro sensor) was used for real-time recording of head LA and AA during collisions and for recording the collision locations <Figure 4>. The sampling frequency of each the three-axis acceleration sensor and three-axis gyro sensor was 1024 Hz (LA of up to 200 G) and 760 Hz (AA of up to 2000 deg/s2) [3]. Measurements were taken during seven games of a regular season of competition. The subjects wore the mouthguard while participating in practices and games. When the mouthguard detected a head movement, it immediately sent a record of the head impact to a software application on a receiving computer. The data were extracted as Microsoft Excel data and analyzed.

Definition of Head Impact Locations

Head impact locations follow those of our previous study [3], according to azimuth and elevation angles within an anatomical coordinate system [13]. All coordinate axes were centered at the head's center of gravity. The positive directions of the x-, y-, and z- axes were defined as follows: (a) x-axis pointing anteriorly toward the face along the midsagittal plane, (b) y-axis directed laterally toward the right ear perpendicular to the midsagittal plane, and (c) z-axis oriented inferiorly toward the trunk along the frontal plane see <Figure 5>.

Impacts were categorized into four head locations see <Figure 5>. The top of the head was defined as the region with an elevation angle (α) greater than 65°, where 0° corresponds to the horizontal plane intersecting the head's center of gravity. The remaining impact locations were categorized based on azimuth angles (θ) as follows: back (−45° to 45°), front (−135° to 135°), and side (including both left and right sides, ±45° to ±135°). Any data points that could not be clearly classified into one of these locations were excluded from the analysis.

Evaluation Items

We defined impact as >10 G of peak LA. We extracted the measurements of the top 50% and top 10% of peak head LA. At those values, we calculated correlation coefficients to examine

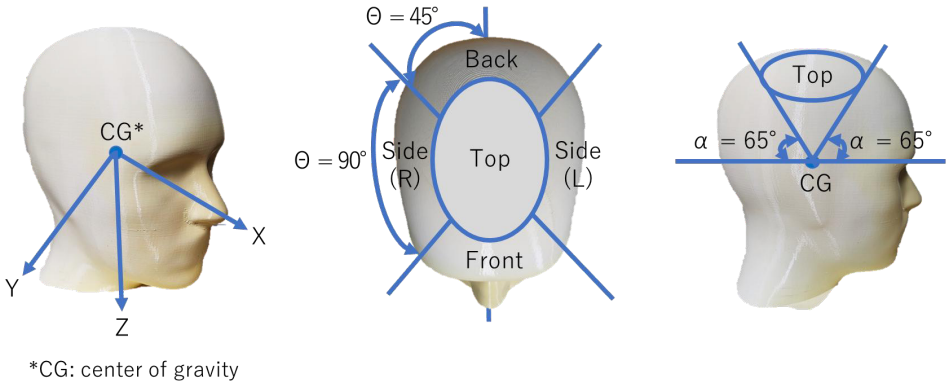

Figure 5. Head impact locations follow those of our previous study, according to azimuth and elevation angles within an anatomical coordinate system.

the relationships between LA and AA, head kinematics and neck strength, head kinematics and neck circumference, and neck strength and neck circumference. In examining the relationship between head kinematics and neck strength, we refer to extensor muscles as "back," flexor muscles as "front," and lateral flexor muscles as "side," in consideration of the collision site and the direction of muscle exertion. At the top, neck strength was excluded because it could not be measured.

Statistical Analysis

We used IBM SPSS version 24 (IBM Japan, Tokyo, Japan) to perform statistical analyses. The significance level was set to 5%. We used Spearman's rank correlation coefficient to investigate the relationships between LA and AA, head kinematics and neck strength, head kinematics and neck circumference, and neck strength and neck circumference. The correlation coefficient was evaluated as follows: very weak (0–0.2), weak (0.2–0.4), moderate (0.4–0.7), or strong (0.7–1.0).

Results

Neck Circumference and Neck Strength

The values of neck circumference and neck strength of the subjects are shown in <Table 1>.

Correlation between LA and AA

Significant positive correlations were observed between LA and AA except for AA at the top 10% of LA (AA@Top10%LA), where the collision site was back (Top50%LA and AA@Top50%LA: ALL; r = 0.467, back; r = 0.315, front; r = 0.501, side; r = 0.496, Top10%LA and AA@Top10%: ALL; r = 0.506, front; r = 0.733, side; r = 0.455; $p < 0.05$) <Table 2>.

Correlation between Head Kinematics and Neck Strength

A significant negative correlation between Top10%LA and neck flexor strength (r = −0.344; $p < 0.05$) was observed, although the correlation was weak. There were significant weak negative correlations between neck flexor strength and AA@Top50%LA and top10%LA (r = −0.230, r = −0.289; $p < 0.05$). A very weak positive correlation was observed between AA@Top50%LA and neck lateral flexor strength (r = 0.121; $p < 0.05$).

Correlation between Head Kinematics and Neck Circumference

Significant, very weak negative correlations were observed between neck circumference and Top50%LA and 10%LA (r = −0.133, r = −0.189, $p < 0.05$). Significant weak negative correlations were observed between neck circumference and AA@Top50%LA and AA@Top10%LA (r = −0.237, r = −0.354, $p < 0.05$).

Correlation between Neck Strength and Neck Circumference

Significant weak and moderate positive correlations, respectively, were observed between neck strength and neck circumference at Top50%LA and Top10%LA (r = 0.242, r = 0.445; $p < 0.05$).

Table 1. Neck circumference and neck strength.

	Neck circumference (cm)	Neck strength (Nm)			
		Flexion	Extension	Side flexion (R)	Side flexion (L)
ALL(n = 22)	40.5 ± 2.3	233.4 ± 92.4	281.8 ± 105.0	298.0 ± 107.2	286.4 ± 104.2

Table 2. Correlation coefficients between head kinematics, neck strength, and neck circumference.

Game (n=7)

	AA @50%LA				Neck strength				Neck circumference
Location	ALL	back	front	side	ALL	back	front	side	ALL
(times)	(1069)	(138)	(303)	(628)	(1069)	(138)	(303)	(628)	(1032)
50% LA	0.467 *	0.315 *	0.501 *	0.496 *	-0.045	-0.102	-0.089	-0.043	-0.133 *
AA @50%LA	-	-	-	-	0.042	0.076	-0.230 *	0.121 *	-0.237 *
Neck strength	-	-	-	-	-	-	-	-	0.242 *

* p<0.05

Game (n=7)

	AA @上位10% LA				Neck strength				Neck circumference
Location	ALL	back	front	side	ALL	back	front	side	ALL
(times)	(199)	(14)	(56)	(129)	(199)	(14)	(56)	(129)	(196)
上位10% LA	0.506 *	0.122	0.733 *	0.455 *	-0.065	0.190	-0.344 *	0.035	-0.189 *
AA @上位10% LA	-	-	-	-	-0.078	-0.060	-0.289 *	-0.099	-0.354 *
Neck strength	-	-	-	-	-	-	-	-	0.445 *

* p<0.05

Discussion

This study aimed to examine whether neck strength contributes to reduced head kinematics during a collision. We found a negative correlation between head LA and AA only in flexion and extension in the sagittal plane, implying that a stronger and thicker neck may reduce head kinematics during a collision.

Correlation between LA and AA

SRC is most likely to result from collisions that involve high levels of head kinematics. In American football players in the USA, head kinematics that leads to SRC is reported to be 69.7–145 G for LA and 7688 rad/s for AA [1,13-16]. We analyzed the collisions in the top 50% and top 10% of LA. We found significant positive correlations between LA and AA except for AA@Top10%LA, where the collision site was back. This result supports previous studies' findings that head kinematics responses during collisions in American football are similar [3,17]. That is, the larger LA acts on a head the larger resultant AA generates on it (and vice versa).

Correlation between Head Kinematics and Neck Strength

When the collision site was front, we found a significant positive correlation between neck strength and Top10%LA. Furthermore, there were significant positive correlations between neck strength and AA@Top10%LA and AA@Top50%LA. These results supported Viano et al. [7], implying that strong neck flexors and extensors may attenuate head kinematics in a collision. Studies in the United States and Japan have reported that high school students receive higher head kinematics during collisions than university students due to weaker neck strength and lack of tackle techniques [18-20].

When the collision site was side, we found, contrary to our expectations, a significant positive correlation between neck strength and AA@50%LA, implying that stronger neck side flexors allow greater rotational motion in the lateral direction.

This is because the cervical vertebrae can move significantly during a side–impact collision; rotational motion occurs simultaneously with lateral flexion, resulting in difficulty in measuring neck rotational strength. [21]. Furthermore, various neck muscles act on rotational motion of the neck as it flexes, extends, and side flexes complicatedly, depending on the location of impact to the head and the direction of force [21,22]. The primary muscle of neck flexion is the sternocleidomastoid, which also acts on neck extension, lateral flexion, and axial rotation [23]. In a future study, it is necessary to clarify which muscles act, and to what extent, according to head impact location [23].

Correlation between Neck Circumference, Neck Strength, and Head Kinematics

We found significant weak and moderate positive correlations between neck strength and neck circumference at Top50%LA and Top10%LA. Additionally, there were significant very weak and weak negative correlations with head kinematics and neck circumference, respectively. These results suggest that a thicker neck equates to greater neck strength and reduced head kinematics (and vice versa). Tierney et al. [6] reported that the head kinematics of female soccer players was ≤44% greater than that of men when striking the ball with the head. It is suggested that this was because women generally have a smaller neck circumference and less muscle strength. Collins et al. [24] reported that low neck strength and small neck circumference correlated with SRC.

A stronger and bigger neck may confer positive psychological protection factors for avoiding injury from intense contact [25]. Although some studies conclude that a strong neck with a large circumference does not provide safety in a collision [5,7,10,11], it is generally recognized as being important for preventing SRC [23].

The neck strength of athletes can be effectively improved by regular neck strength training [2,26], but not their neuromuscular response [11]. Peak head kinematics normally occurs at 10–20 ms after impact [10,18]; however, voluntary contraction of the neck muscles occurs 65–90 ms after muscle activation [27]. It is important to increase the rigidity of the head and neck before a collision by pretensioning the neck muscles [6,10]. Even if there is high peak muscle strength, stiffness of the head and neck may not be sufficient if pretensioning of the neck muscles is not exerted in time [23]. In fact, many studies have reported that the timing of this pretensioning reaction is more important than peak muscle strength at the time of collision [6,10]. Additionally, players who had SRC history may have delayed onset of muscle activation although we did not consider SRC history due to limited subjects.

Limitations

Our study has several limitations. First, we together analyzed data obtained during preseason training as well as the competition season. Second, in measuring neck circumference, we did not quantify muscle volume; thus, we cannot equate greater neck circumference with larger muscle volume. Finally, our statistical analyses derived a correlation coefficient that does not clarify causal relationships between head kinematics, neck strength, and neck circumference.

Conclusions

Our findings point to the need to develop effective neck strength training programs, especially for American football players. In consideration of the limitations of our study, future study into the mechanics of SRC is warranted.

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Conflicts of Interest

The authors declare no conflict of interest.

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